

1. **Generalized eigenstates.** [*Numerical, TISE code*] In this project, you will compute the properties of the generalized eigenstates $\psi_E(x)$ of a particle in 1D, subject to the potential

$$V(x) = \begin{cases} \infty & x < 0 \\ V_0 \cos^2 \left[\frac{\pi(x-x_0)}{2\sigma} \right] & x \in (x_0 - \sigma, x_0 + \sigma) \quad (x_0 > \sigma > 0, \quad V_0 > 0) \\ 0 & \text{otherwise} \end{cases}$$

First, adapt the TISE program to compute the eigenstates $\varphi_n(x)$ and eigenvalues E_n of the particle confined to an interval $[0, R]$, with $R \geq x_0 + \sigma$ and with zero boundary conditions at $x = 0$ and $x = R$, in the finite basis spanned by the B-splines from the second, to the one next to last, $\{|B_i\rangle, i = 2, \dots, N-1\}$, and with normalization $\langle \varphi_n | \varphi_{n'} \rangle = \delta_{nn'}$. Thus, $n = 1, \dots, N-2$. In this way, you will find those generalized eigenfunctions that happen vanish at $x_0 + \sigma$, $\varphi_n(x) \propto \psi_{E_n}(x)$ (beyond a certain energy, the Bsplines are not able to accurately reproduce the exact solutions). Now, we need to figure out how to find the generalized eigenfunctions for arbitrary values of the energy E . For $x > x_0 + \sigma$, the generalized eigenfunctions have the asymptotic form

$$\psi_E(x) = \sqrt{\frac{2}{\pi k}} \sin[kx + \delta(E)], \quad k = \sqrt{2E},$$

where the prefactor $\sqrt{2/\pi k}$ ensures the normalization $\langle \psi_E | \psi_{E'} \rangle = \delta(E - E')$. In general, therefore, these functions do not vanish at $x_0 + \sigma$. Hence, they are proportional to a linear combination of the confined eigenstates, plus the last B-spline, which does not vanish at the boundary

$$|\psi_E\rangle = A_E |\bar{\psi}_E\rangle, \quad |\bar{\psi}_E\rangle = \left[\sum_{n=1}^{N-2} |\varphi_n\rangle c_n + |B_N\rangle \right].$$

If these are to be eigenfunctions of H , it must be $(E - H)|\psi_E\rangle = 0$, and in particular $\langle \varphi_n | (E - H) | \psi_E \rangle = 0 \quad \forall n \in I_{N-2}$. By imposing the latter condition, we immediately find the expression of the coefficients

$$c_n = \frac{\langle \varphi_n | H | B_N \rangle - E \langle \varphi_n | B_N \rangle}{E - E_n}.$$

These coefficients determine the generalized eigenstate up to a normalization factor. To determine the normalization factor and the value of the phase shift $\delta(E)$, we observe that that

$$\psi_E(R) = \sqrt{\frac{2}{\pi k}} \sin(kR + \delta), \quad \psi'_E(R) = \sqrt{\frac{2k}{\pi}} \cos(kR + \delta)$$

and hence

$$k \psi_E^2(R) + \psi_E'^2(R)/k = \frac{2}{\pi}, \quad \delta(E) = \arctan \left[\frac{k \psi_E(R)}{\psi_E'(R)} \right] - kR$$

which implies

$$A_E = \sqrt{\frac{2/\pi}{k \bar{\psi}_E^2(R) + \bar{\psi}_E'^2(R)/k}}, \quad \delta(E) = \arctan \left[\frac{k \bar{\psi}_E(R)}{\bar{\psi}_E'(R)} \right] - kR.$$

Include in the program a loop over a uniformly spaced grid of energies $\varepsilon_i = \frac{E_{\max}}{N_E} i$. For each energy, compute the corresponding generalized eigenfunction ψ_{ε_i} , tabulate it on a grid $x_i = R/(N_x - 1)(i - 1)$, and save it to a file (two columns: one for the x coordinate, and one for $\psi_{\varepsilon_i}(x)$). Save to a file also the phaseshift as a function of the energy, as well as its derivative, computed as $d\delta/dE = \frac{1}{2 \cos(2\delta)} d \sin(2\delta)/dE$ (in this way, one avoids the discontinuities of $\delta(E)$). Plot $d\delta/dE$ as a function of the energy, identify energies of interest, and plot the corresponding wavefunctions.